OBJECTIVES

BIO201 FINAL TERM- 100%

2023 CELL BIOLOGY

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- The signal sequences for the action of proteins is present in -------? ***Polypeptide chains***
- Proteins sequences begin on -------? ***Free ribosomes***
- An NLS is necessary for import of proteins into the ------? ***Nucleus***
- Which mutation passes on the daughter cells after the mitosis? ***Somatic mutation***
- Point mutation has base pair? ***Single***
- No charge in amino acid sequence in ------? ***Silent mutation***
- A protein that carries oxygen in humans? ***Red blood cells***
- ------- is the study of the heritable changes in gene expression? ***Epigenetics***
- Cluster of genes with ------ promoter is called operon? ***1***
- Tryptophan blocks RNA polymerase from ------? ***Binding and transcribing genes***
- Which regulatory molecule enhances transcription? ***CRP-cAMP***
- An operon is regulated by ------? ***Activated protein***
- DNA is highly ordered around positively charged -----? ***Histone proteins***
- The fusion of membranes from the same compartments is termed as ---- ***Homontypic fusion***
- Which substance in immune system causes capillaries to dilate and become leaky? ***Histamine***
- On which site a substrate binds on an enzyme----- ***Active site***
- For optimum activity of hydrolytic enzymes, lysosomes maintain a pH about. ***5.0***
- Which one of the following helps in stimulating breast cells to produce milk? ***Prolactin***
- Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it ------ ***Water soluble proteins***
- Activation energy of an enzymatic reaction is ***Low***
- Oncogenes after mutation produce one of abnormal condition. ***Cancer***
- Philadelphia chromosome is a chromosomal translocation between chromosome ---- and ---- ***9 & 22***
- Cyanobacteria is a type of bacteria that has---- ***Chlorophyll***
- Helices that are involved in against binding are ***Helix 5***
- The lipids molecules held together by weak interactions and ----- ***Van der Waals forces***
- Beta blockers are involved in ***mimicking the function of a hormone.***
- Initiation site of transcription ***Promoter***
- Sex related disorders are ***X- related***
- Function of G protein in vision may ---- ***Activation of Phospholipase c- Beta***

- Ran GTpase is a molecular switch that exits in **2 states**
- Dendritic cells use on which of the following signal to guide helper T cells to effector T cells ------- ***IL- 2***
- TH2 cells mainly help to activate: ***Bcells***
- Fibroblast cells can be induced to proliferate by treatment with.............. ***Epidermal growth factor (EG)***
- Even in the absence of antigen humans can make more than..........***different antibody molecules. 1012***
- The_____ immune system, like the____ system, can remember prior experiences ***Adaptive, nervous***
- Inhibitors that binds to a site other than active site of enzyme. ***Irreversible inhibitor***
- Condition in which cells have specific binding. ***Cell adhesion***
- People with the Y chromosome have external appearance. ***Abnormal***
- Which one is the product of genetic engineering? ***Inulin***
- Which one is initiation codon? ***AUG***
- Function of Golgi apparatus is ------ ***Packing and transportation of proteins***
- An average globular cell has ----- mitochondria. ***10,000***
- Chromosome fibers consist of---- ***DNA & proteins***
- Microtubules which are not attached to the chromosome are----- ***Mitotic center***
- The number of appearance of chromosomes in the nucleus of a eukaryotic cell is called--- ***Karyotype***
- HIV is a --- ***RNA virus***
- Ribosomes are engaged in ---- ***protein synthesis***
- ------ are removed during RNA maturation. ***Introns***
- Mendel donated dominant and recessive traits as ----- ***SS and ss***
- Structure of protein is ---- ***three dimensional***
- The small subunits of ribosomes are bind to mRNA this sequence is called? ***ShineDalgarno sequence***
- After translation, ---- can be removed? ***Methionine***
- rRNA in small subunit stabilizes ----- between m and rRNA? ***H- binding***
- Large subunits break bond between? ***tRNA***
- Protein folding information is a function of ----- sequences? ***AA***
- Which groups reduces positive charge when add to histones? Acetyl Promoter has ----- essential sequences? ***Two***
- TATA box where ---- begins to denature? ***DNA***
- Eukaryotic promoter have TATA box, which is located----? ***25bp***
- The first transcription factor is -----? ***TF11D***
- Many genes have enhancer away ---- nucleotide? ***1000***
- During RNA maturation introns are ----? ***Removed***
- RNA has complimentary bases that bind CS at ------ exon- introns boundary? ***5'***
- The active X makes RNAi anti Xist gene approximately called ----? ***Tsix***
- One receptor----- active Gs? ***100***
- Per receptor-hormone complex are produced by ----- cAMP? ***Hundred***
- Release of glucose fuels ----- response? ***Flight or fight***
- AKA cAMP depend protein -----? ***Kinase***
- The PKAs are -----? ***Tetramerase***
- PDE1 receptor interacts with -----? ***G1***
- The G protein the synthesis of cAMP by -----? ***Adenylyl cyclase***
- One rhodopsin molecule absorbs ----- proton? ***One***
- Calmodulin binds 4 Ca 2+ allosteric (PKA) ----? ***1%***

- Tired brain produces-----? Adenosine
- Extracellular ligands binding domain on ----- segments? ***1 trans membrane***
- Most numerous receptors tyrosine ------? ***Kinases***
- EGF activated MAPK stays active for ---- minute? ***5***
- ---- Genes encode kinases? ***2%***
- ----- Proteins are phosphorylated? ***1/3***
- Activated RAF is phosphorylates many molecules of -----? ***MEK***
- Which attract chromosomal proteins? ***Methylation & Histone DE acetylation***
- Multicellular organisms cells are specialized for the support ----- / ***Brain sugar***
- Our sense organs allow us to chemical respond to -----? ***Taste***
- Signal detection to final response is called---? ***Signal transduction pathway***
- Which signal receptors on the cells that secrete them? ***Autocrine signal***
- Cell behavior in an environment flooded with ------ of Ligands ***Hundred***
- Inactive ICR may be ----- bond? ***DNA***
- How many types of Proteins? ***8***
- Wich protin spread signal from one pathway to another? ***Bifurcation***
- Which protein produce intracellular mediatores? ***Amplifier***
- Many cells use --- proteins to enhance porteins? ***Scaffold***
- ----- are produced by in response to signal received by the receptor? ***SCIMs***
- Helices ------ that participate in against binding? ***3,5 & 6***
- Ligand activates --- proteins? ***G***
- cAMP can bind ----- to open them? ***Ion channels***
- APC gene mutation occurs in human colon cancer -----? ***80%***
- Proteolysis means protein ------? ***Breaking***
- APC adheres junctions scaffold protein called -----? ***Axin***
- BTK and PLC –Y to the cytoplasmic face of the -----? ***Plasma membrane***
- ABL protein preventing ----? ***Binding***
- Stimulation of milk production is due to -----? ***Prolaction***
- Phagocytes engulf ---- and dead cells? ***Bactria***
- Erythrocytes transport oxygen and ---? ***Carbon dioxide***
- Which cells release histamine when damaged? ***Mast cells***
- T cells mature in ----? ***Thymus***
- Lymph nodes contain ----? ***WBCs***
- AIS have ---- components? ***2***
- B lymphocytes kill pathogens ----? ***20%***
- How many lymphocytes in human body? ***2×1012***
- Substances that activate AIS are called ----? ***Antigens***
- The rates were unable to mount? ***AIR***
- B cells develop in ----? Bone marrow
- Immunoglobin domain is ---- long? ***110 AA***
- Polyvalent ---- times stronger than monovalent? ***100***
- Which is the first class of Ab to appear on the cells surface? ***lgM***
- How many classes of antibodies? ***5***
- Which chain is required for pen tamer formation? ***J chain***
- lgG is monomer it is only -----? ***Ab***
- Which receptors are located on the mast cells? ***lgE Fc***
- Bruton's tyrosine mutation is associated protein kinase B? ***X linked*** Which signal cells indirectly activating protein kinase B? ***Pl 3***
- Intracellular protease cleaves off the cytoplasmic tail of ----- when it binds ----? ***Notch, delta***

- Activated protein is -----? ***PKB***
- Which receptor required for signaling? ***Type 1- TGF beta & Type 2 TGF beta, Both***
- Smad complex move into ------- for association of other gene regulatory proteins? ***Nucleus***
- ----- Protein called Disheveled is activated when Wnt is present. ***Cytoplasmic signaling***
- Which is the primary source due to the skin is lost? ***Injury infection***
- Pathogens may cause infection of ----? ***Digestive system***
- Our tears and saliva have enzymes that destroy ----? ***Microorganisms***
- Steps of immune system? ***4***
- Histamine diffuses into ----? ***Capillaries***
- A myeloma cell is fused with a B cells to form? ***Hybridoma***
- In the absence of the antigen human can make more than ---- different antibody molecule? ***1012***
- How many types of MHC molecules? ***2***
- APC expresses major histocompatibility complex -----? ***MHC genes***
- How many genes are present in MHC classes? ***6***
- 3 genes for class 1 proteins -----? ***HLA-A,B,C***
- Class 11 can ------pair up? ***α & β***
- MHC rove can accommodate an extended peptide about ----- long? ***8-10 amino acids***
- Each class 1 MHC protein can bind a peptide of -----? ***Diverse sequences***
- Which terminal bonds to invariant pockets? ***C , N***
- Cytotoxic T cells provide protection against -----? ***Intracellular pathogen***
- Which cell form a clone? ***Tc***
- How many steps are present in protein depending killing? ***4***
- Helper T cells activated by an ---- cell to become an affecter cell? ***Antigen cell***
- How many types of effector cells? ***2***
- TH2 cells mainly help activate -------? ***B cells***
- APCs provide ---- types of signals? ***2***
- Signal 1 is produced by ---? ***Peptide bond***
- B 7 proteins are -----? ***CD80- Cd86***
- Which protein is present on the surface of T cells? ***CD28***
- The compliment system consists of ----- proteins? ***30***
- The complement proteins are account for about ----- of the globulin? ***5%***
- C1 complex consists of ---- molecules? ***6***
- Which molecule binding the surface of pathogens? ***C1q***
- The complex of C4b, 2b is called -----? ***C3***
- Macrophages and neutrophils have receptor for -----? ***C3b***
- Cleavage of CS by the -----? ***C3/C***
- C3b binds to a protein is called? ***Factor B***
- C3 converts into ----- form? ***3***
- The MBL associated serine proteases-----? ***MASP-1, MASP-2, MASp-3***
- Factor 1 is inactivates -----? ***C3b***
- C1 inhibitor binds to sites on activated -----? ***C11NH***
- A proto- oncogene code for -----? ***Protein***
- Proto- oncogenes are involved in signal ------? ***Transduction***
- How many classes of oncogenes? ***4***

- Autonomous cell growth is present in ----- cell? ***Cancer***
- Many glioblastomas secrete -----? ***PDGF***
- Homotypic fusion and heterotypic fusion requires a set of matching -----?
- ***SNAREs***
- Resident ER membrane protein consists of ----- lysine? ***2***
- Lysosomes contain ------ types of hydrolytic enzymes? ***40***
- Lysosome contains pH ----? ***5.0***
- Lysosomes hydrolysis carries a unique marker in the form of ----- groups? ***M6P*** How many classes of WBCs in mammals? ***3***
- All organs have limited ability to regenerate ----? ***1cm***
- Human blood discovered? ***Austrian Karl Landsteiner***
- Phagocytosis involves the digestion of ----? ***Microorganisms***
- Skin regenerate every -------? ***two weeks*** Every ---- a person dies from a disease that can be treated with tissue transplant?
- ***30 sec***
- In ----- Bogdanov blood transfusion experiment? ***1924***
- Which cells don't cause e rejection? ***Adult stem cells***
- EGF receptor ERBB1 is over expressed in ------ of SCC of lung? ***80%***
- ERBB/HER2 is amplified in the ------ of the cancer? ***25%-30%***
- Mutation of RAS is present approximately -----in all humans? ***30%***
- BCR-ABL hybrid gene activates -----? ***RAC***
- MYC assembles own a specific DNA sequence and drive ----- of genes? ***Transcription***
- MYC binds with -----? ***DNA***
- MYC gene encoding the -----? ***Cyclin D***
- RB which overcome the hurdle of ------ hurdle? ***G1 to 5***
- The activated of CDKs is regulated by ------? ***CDK***
- CDKs are occurring in ----? ***TGF-α***
- Which can activate DNA repair protein? ***P53***
- Which is the most common mutant gene in human cancers? ***P53***
- An animal cell contains about ------- protein molecule? ***1010***
- How many kinds of protein molecules? ***10,000-20,000***
- Lipid bilayer of organelle membranes is impermeable to most ---- molecule? ***Hydrophilic***
- The transport molecule derived from the lumen of ----- compartment? ***1***
- ---- direct the protein form the cytosol in to ER? ***SS1***
- SS2 consist of a specific -------dimensional arrangement of atoms on the protein surface? ***3***
- Envelop consist of ----- concentric membrane? ***2***
- The inner nuclear membrane contains specific ----- that acts as a binding site? ***Protein***
- Bidirectional traffic between the cytosol and -----? ***Protein***
- Each complex has a molecular mass of about -----? ***125 million***
- THE SIGNALDEFINED BY NUMEROUS PROTEIN BY USING ----- TECHNOLOGY? ***DNA recombinant***
- The energy is provided by the hydrolysis of -----? ***GTP***
- TIM22 complex mediates ------- of inner membrane protein? ***Insertion***
- The N- terminal signal sequence of precursor protein is recognized by ------ complex? ***TOM***
- Protein transport requires ----- signal sequences? ***2***
- How many sub compartments of mitochondria? ***2***

- Chaperon protein belonging to the ----- family? ***hsp70***
- The trans locator ----- complex forms a water filled pore in the membrane? ***Sec61***
- The stop transfer sequence is a ------ α helix membrane? ***Single***
- Golgi apparatus are packed into a small ----- coated transport vesicle? ***COP11***
- P53 protein is stabilized by ------? ***Damaged DNA***
- How many types of Cdks? ***4***
- Active P53 binds to regulatory region of ------ gene? ***P21***
- Abnormally high level of MYC causes the activation of -----? ***P19ARF***
- Papillomavirus uses ----- viral proteins? ***2***
- Two vital proteins are ----? ***E6 and E7***
- Which virus uses a single dual purpose protein? ***Sv40***
- How many bone cells die in a healthy adult? ***Billions***
- ------ Biochemical processes take place in a membrane? ***Vital***
- Number of Human gene is ----? ***30,000***
- Immunoglobin genes have relatively ----- mutation rate? ***High***
- Effector helper T cells stimulate the responses of other cells? ***Phagocytic macrophages, B cells***
- T cells receptor recognizes processed antigen of other cells? ***MHC protein***
- How many proteins are not immunoglobin? ***1***
- Antigen is -----? ***Peptide fragment***
- A virus infected cell in the case of -----? ***Cytotoxic C cell***